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Effect of gap distance on vortex-induced vibration in two parallel cable-stayed bridges☆

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ABSTRACT

This study was focused on the vortex-induced vibration (VIV) in parallel-disposed twin cable-stayed bridges. A series of wind tunnel tests were carried out to investigate the gap distance between two bridge decks and its effect on VIV. The gap distances examined covered a wide range of from one to twenty times the depth of the upstream deck. The amplitudes of VIVs in both decks were estimated using the variations in wind velocity. Particle image velocimetry (PIV) tests were performed to identify streamline changes between two decks for different gap distances. Also, the distribution of the swirling strength of the generated vortices was evaluated based on the velocity data obtained from PIV testing. According to the research results, gap distances of five to seven times the depth of the upstream deck critically affected the interactive VIV in two parallel cable-stayed bridges. The vulnerable gap distance was closely related to the moving distance of one vortex during a period of oscillation of the deck.

1. Introduction

Vortex-induced vibration (VIV) is an important issue in the aerodynamic performance of parallel bridges, and is typically connected to fatigue damage and to the serviceability of a bridge. Even though, several researches (Kimura et al., 2008; Seo et al., 2013; Kim et al., 2013; Argentini et al., 2015) have reported VIV characteristics in parallel bridges, many things are yet to be explained due to the complexity of structure-flow interactions. One of these unexplained factors is the gap distance. The gap distance is generally known to be a key parameter for two in-line bodies in a flow situation. Kimura et al. (2008) demonstrated that the gap distance between a pair of parallel bridges is a critical factor in the response characteristics of VIV; nonetheless, a general conclusion was not stated.

The target bridges in this study are those composed of a pair of cable-stayed bridges with decks that are similar in shape. VIV was observed in an upstream deck in 2011. The observed VIV exceeded a serviceability criterion of 0.5 m/s^2. This was the first observation of the VIV problem in a parallel bridge while in service. Seo et al. (2013) investigated the vulnerability of an upstream deck to VIV through a series of wind tunnel tests. The successful reproduction of observed VIV in a wind tunnel demonstrated that a single bridge alone could perform satisfactorily, but the VIV was magnified by the parallel arrangement of two decks. Particle image velocimetry (PIV) tests identified well-developed motion-induced shedding vortices in the gap area between two decks. In a follow-up study, Kim et al. (2013) successfully demonstrated the existence of interactive VIV, predicted by wind tunnel tests, from the field monitoring data of the bridges. Both studies reported low-level inherent damping ratios from upstream bridges and a potential contribution to the observation of higher-level vibrations.

Two previous studies that were focused on bridge aerodynamics actually reported VIV in 2011, but these studies were limited to the fixed gap distance of existing bridges. These two previous papers helped point out the current state of the target bridges, and an extension of the investigation on various gap distances could be indispensable for deeper insight into this complicated phenomenon. Study of the gap distance should help evaluate how the current gap distance of the target bridges contributed to the VIV in 2011.

In order to build on this research background, this study investigated the effect of gap distance on the VIV of target bridges using a series of wind tunnel tests. A PIV was also designed to visualize the changes in the flow patterns of the gap for various gap distances.

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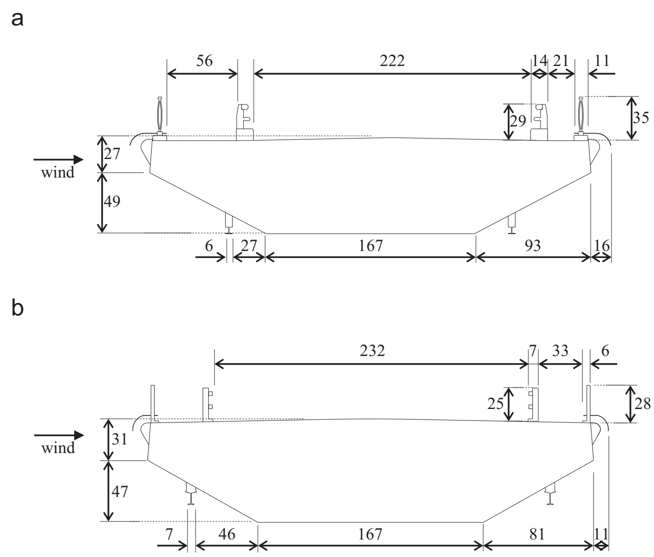

Fig. 1. Dimension (mm) of section models: (a) B2 and (b) B1.

Table 1
Similarity scales and setup parameters for sectional models.

Parameters	Similarity scale	B2		B1	
		Prototype	Model	Prototype	Model
Length (m)	λ_L=1/36	32.400	0.900	32.400	0.900
Breadth (m)	λ_L	12.690	0.353	11.860	0.329
Depth (m)	λ_L	2.750	0.076	2.800	0.078
Aspect ratio	1	4.615	4.645	4.236	4.218
Mass (kg/m)	λ_L^2	8978	6.874	6950	5.400
Vertical frequency (Hz)	λ_f=11.7	0.436	5.096	0.513	5.875
Vertical damping ratio (%)	1	0.2–0.3	0.1	0.5–0.7	0.1
Scruton number	1	23.9–35.8	12.0	44.6–62.4	8.9

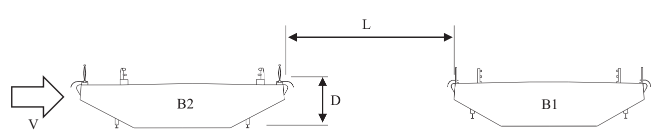

Fig. 2. Nomenclature of wind tunnel tests.

2. Setup of wind tunnel tests

Two-dimensional section model tests were carried out in the wind tunnel located in the Department of Civil and Environmental Engineering at Seoul National University, Korea. The test section of the wind tunnel is 1.5 m in height, 1.0 m in width and 4.0 m in length. The maximum wind velocity of the wind tunnel is 23 m/s.

The two section models were identical to those used in the studies by Seo et al. (2013) and Kim et al. (2013). The geometries and dimensions of the sections are shown in Fig. 1. The upstream deck is referred to as B2 — this resembles the one that showed the VIV problem in 2011 — and the downstream deck will be referred to as B1. The section models were manufactured with the length scale $\lambda_L = L_m/L_p = 1/36$, where L_m and L_p represent the length of the model and prototype, respectively.

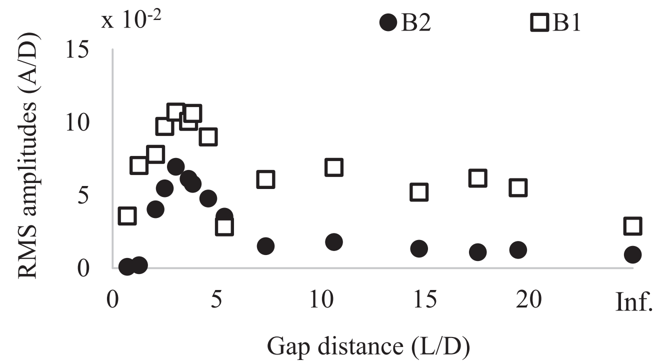

Fig. 4. The maximum single RMS amplitudes for each gap distance.

For the dynamic tests, both section models were separately mounted on spring-supported systems that permitted both vertical and torsional motions of the models. The setup parameters for dynamic tests are listed in Table 1. The frequency scale was set to $\lambda_f = f_m/f_p = 11.7$, where f_m and f_p were the frequencies of the model and the prototype, respectively. The damping ratios for the vertical mode were set to a minimum level of 0.1% for better observation of the vibrations. The Scruton numbers of both the models are defined by the depths of each deck-section model.

In the present study, the gap distance between the two section models and the wind velocity were considered to be experimental variables. The gap distance and wind velocity are expressed in non-dimensional forms as L/D and V/nD, where V is the upcoming wind velocity, L is the gap distance between two decks, D is the depth of the upstream deck, and n is the motional frequency of the upstream deck, as shown in Fig. 2.

The vibration responses were intensively investigated for L/D=0.7, 1.3, 2.1, 2.5, 3.0, 3.6, 3.9, 4.6, 5.4, 7.4, 10.6, 14.7, 17.6, 19.5, and ∞. The infinity value denotes a single deck case. The PIV tests were conducted for L/D=2.5, 3.0, 3.6, 3.9, 4.6, 7.4, and 11.3. The experimental setup is shown in Fig. 3. The vertical and torsional displacements of two decks were measured using a total of 8 (4 for each section model) laser displacement transducers with a sampling frequency of 500 Hz. The upcoming wind velocity was measured with a pitot tube located in the front of the decks. A high-resolution PIV system was composed of a high-resolution, two-mega-pixel CCD camera, a double-pulsed ND: YAG laser device emitting two pulses of 135 mJ at the wavelength of 532 nm, a seed generator spreading fine olive oil droplets of 1 μm, and a trigger generator.

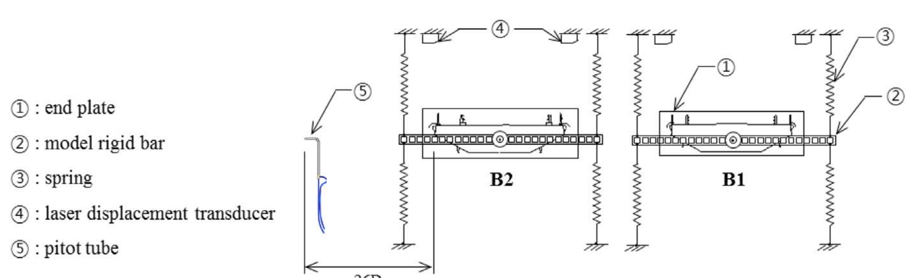

Fig. 3. Experimental setup.

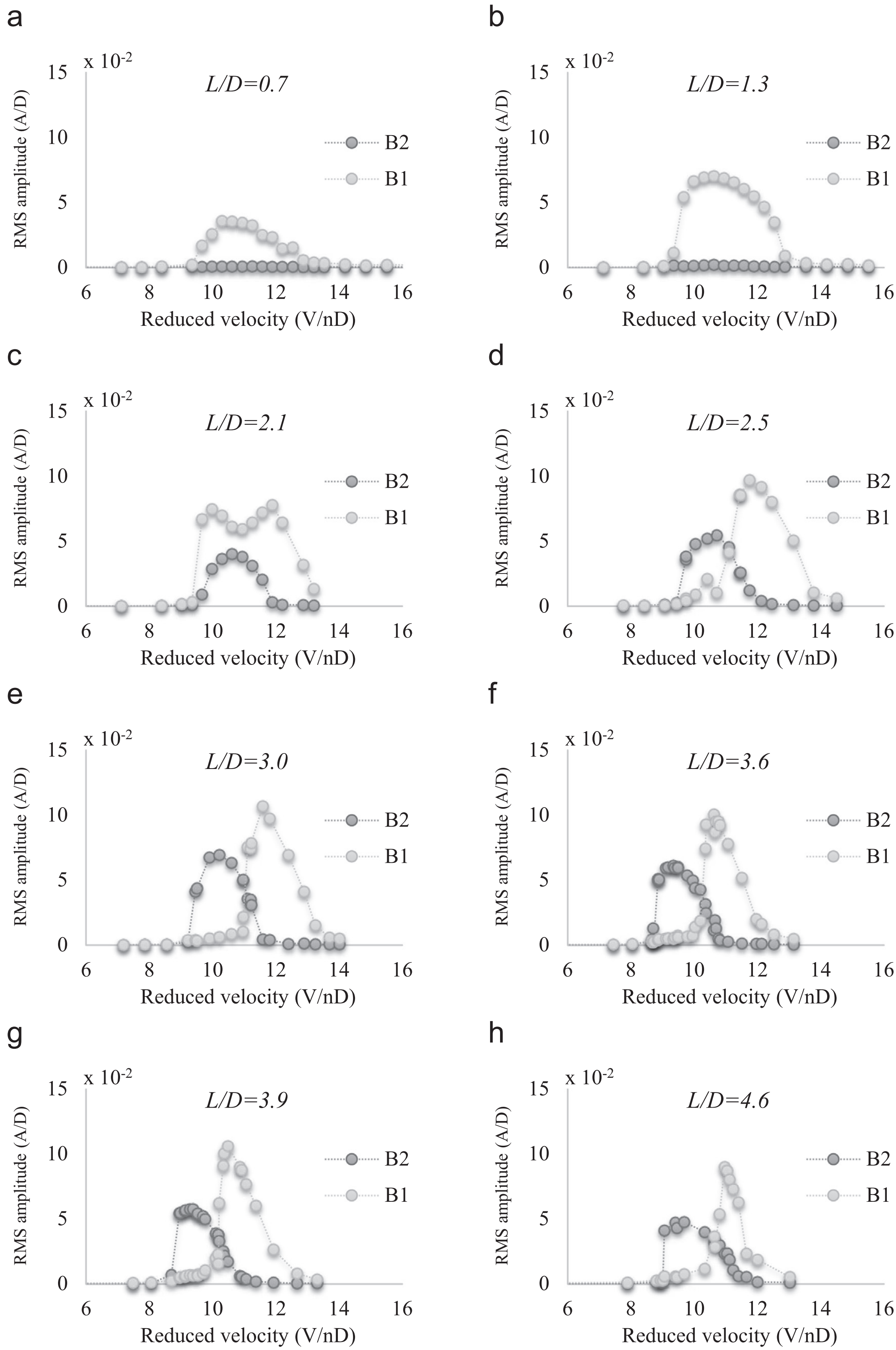

Fig. 5. A-V curves for the gap distances of (a) 0.7, (b) 1.3, (c) 2.1, (d) 2.5, (e) 3.0, (f) 3.6, (g) 3.9, (h) 4.6, (i) 5.4, (j) 7.4, (k) 10.6, (l) 14.7, (m) 17.6, (n) 19.5 and (o) infinity. ,.

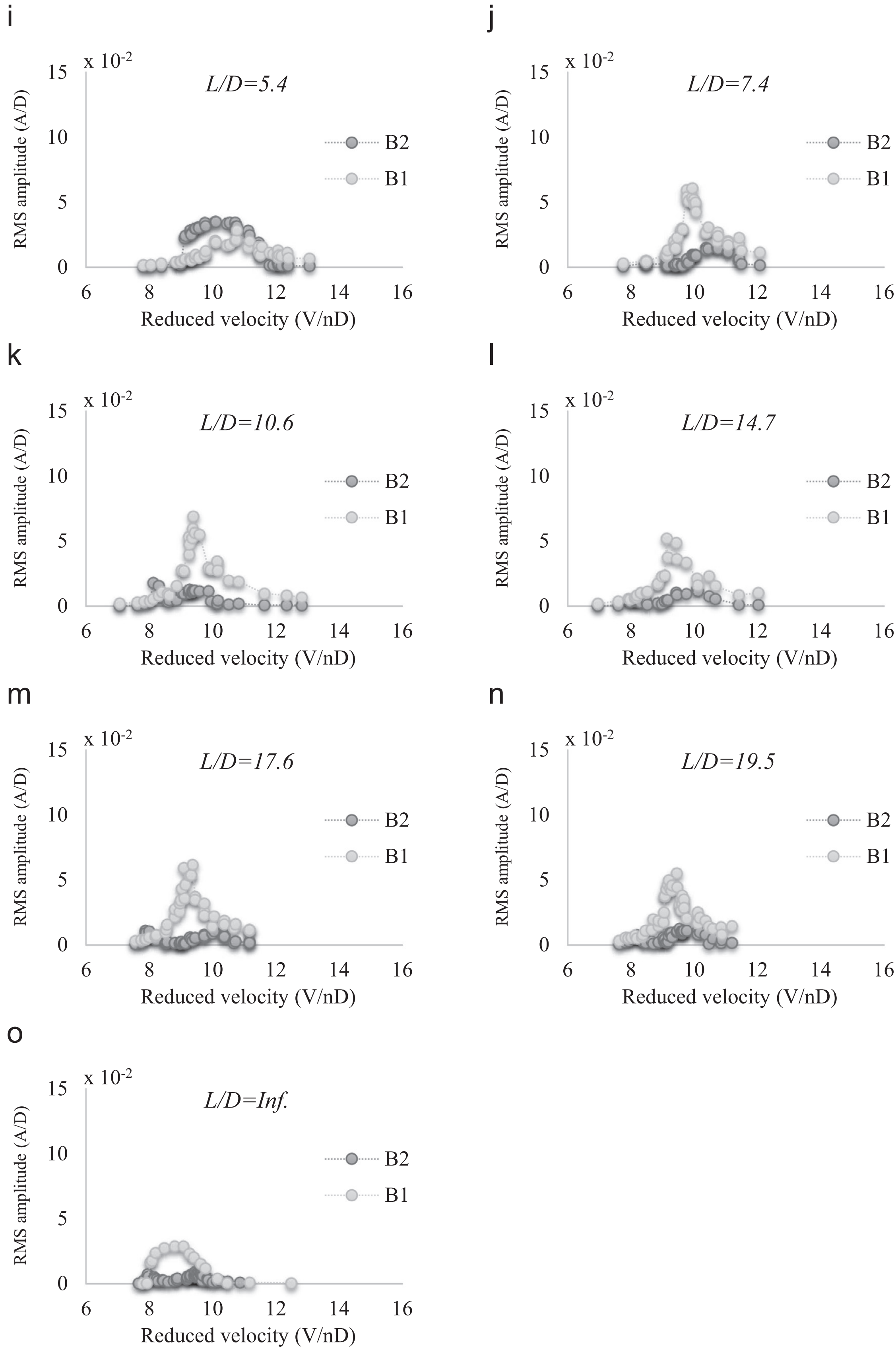

Fig. 5. (*continued*)

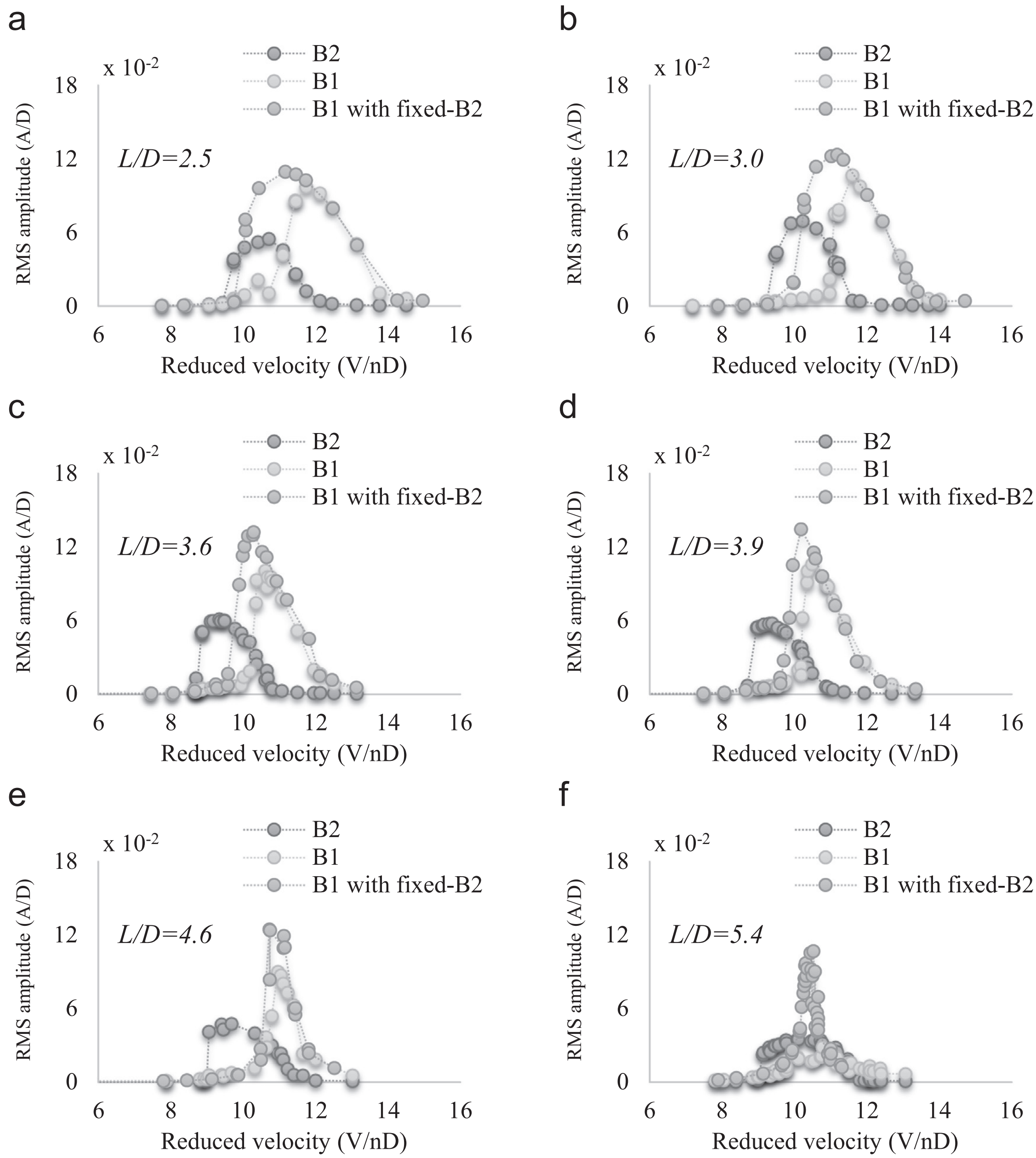

Fig. 6. A-V curves of B1 while the motion of B2 is being restrained or released at gap distances of (a) 2.5, (b) 3.0, (c) 3.6, (d) 3.9, (e) 4.6 and (f) 5.4.

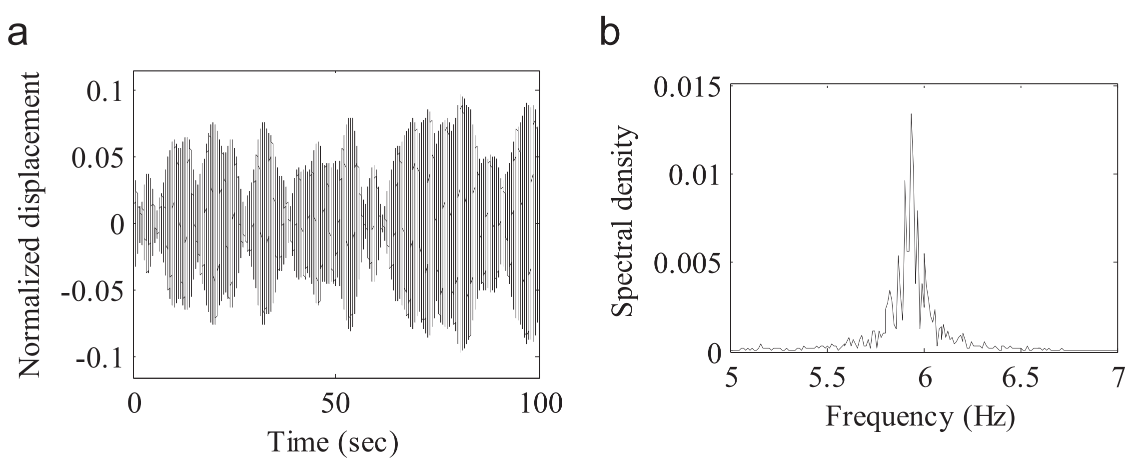

Fig. 7. (a) Time history of normalized vertical displacement of B1 and (b) its spectral density at L/D=7.4 and V/nD=10.3.

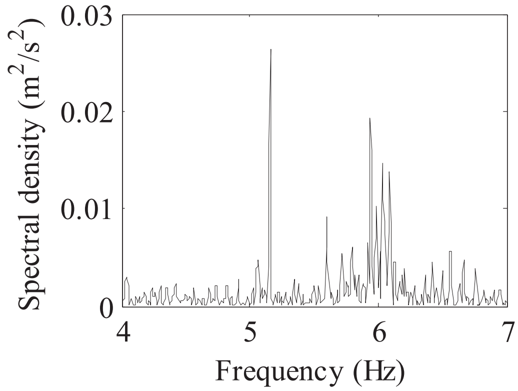

Fig. 8. Spectral density of vertical wind velocity fluctuation measured behind B2 at L/D=7.4 and V/nD=10.3.

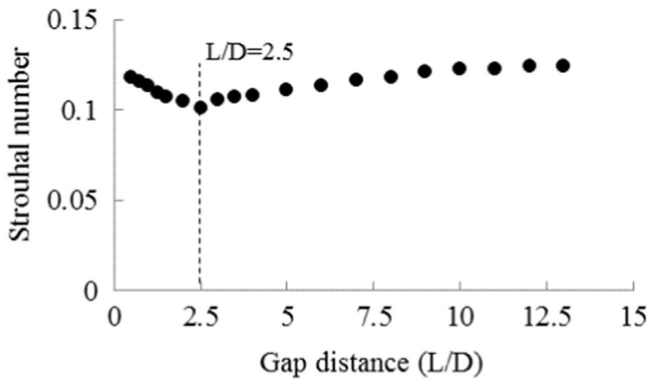

Fig. 9. Strouhal numbers according to the gap distance.

3. Influence of gap distances on the VIVs of parallel bridges

A series of wind tunnel tests were performed to observe the effects of gap distance on VIVs under smooth flow conditions. A total of 15 gap distances were considered including L/D=3.6, which corresponded to the gap distance of the as-built bridges. Since no meaningful torsional oscillation was observed during testing, only vertical responses were taken into consideration in further discussions. For a non-dimensional representation, RMS responses of a single amplitude (A) for both decks were divided by the depth (D) of B2.

The maximum RMS single amplitudes were obtained during fully developed VIVs in both decks for each gap distance, and the results are summarized in Fig. 4. Seo et al. (2013) reported that the VIV of B2 was amplified by the parallel arrangement of the two decks, and this was based on the comparative wind tunnel tests between parallel and single-deck arrangements with the fixed-gap distance of as-built bridges. The same pattern of amplified VIVs is identified in Fig. 4 for the gap distances between L/D=2.1 and 5.4, even though the level of amplification is dependent on the gap distance. The amplification in the magnitude of VIV was observed for the gap distances in both the windward deck (B2) and the leeward deck (B1). In fact, the amplitudes of B1 generally exceeded those of B2. The peak amplitude of VIVs appeared at the gap distance of L/D=3.0, which approximates that of as-built bridges. The peak amplitudes in both bridges decreased as the gap distance increased to L/D=5.4. When the gap distance exceeded a value of L/D=7.4, the amplitudes of VIV in both decks were only marginally affected.

Kimura et al. (2008) also investigated the vulnerable gap distance of VIV for gap distances of 2–8 times the deck width, as measured center-to-center, which were equivalent to 6–24 times the depth of the deck, and reported interference effects of gap distance that were 8 times the deck widths. We, however, do not recommend a direct comparison of their results with this study, because the differences in the setup frequencies and in the bluffness of the parallel decks in the two studies could seriously affect both the interference phenomena and the effect of the gap distance.

3.1. Interactive VIVs for the gap distances of L/D=0.7–1.3

The amplitude-velocity (A-V) curves, which demonstrate the relationship between RMS single amplitudes of B1 and B2 to wind velocity, are shown in Fig. 5, covering all gap distances considered. As shown in Fig. 5(a) and (b), VIV was observable in only the leeward deck (B1) for narrow gap distances of L/D=0.7 and 1.3. The reduced lock-in wind velocity ranged from V/nD=9–13, and the magnitude of the VIV was amplified when compared with that for the single deck in Fig. 5(o). It was noteworthy that the reduced velocity of 10 corresponded to a wind velocity of about 12 m/s in the prototype scale. A gap distance between 0.7 and 1.3 was not wide enough to allow the formation of vortex trails behind the windward deck.

3.2. Interactive VIVs for the gap distances of L/D=2.1–5.4

As the gap distance reached L/D=2.1, VIV began in the windward deck (B2), as shown in Fig. 5(c). Further increasing the gap distance led to a VIV in B2 that reached the peak amplitude at a gap distance of L/D=3.0, as shown in Fig. 5(e).

Even though the magnitude of the VIV in the leeward deck (B1) was also increased up to a gap distance of L/D=3.0, the lock-in wind velocity of B1 was affected by the VIV motion in the windward deck. As shown in Fig. 5(c), both decks showed VIV at the same reduced wind velocity of 9.0. The range of the lock-in wind velocity of B2 was within that of B1. The amplitude of B1 was decreased according to the increase in the amplitude of B2 at a reduced wind velocity of 11.0. For the gap distances of $2.5 \leq L/D \leq 4.6$ (see Fig. 5(d)-(h)), B1 showed

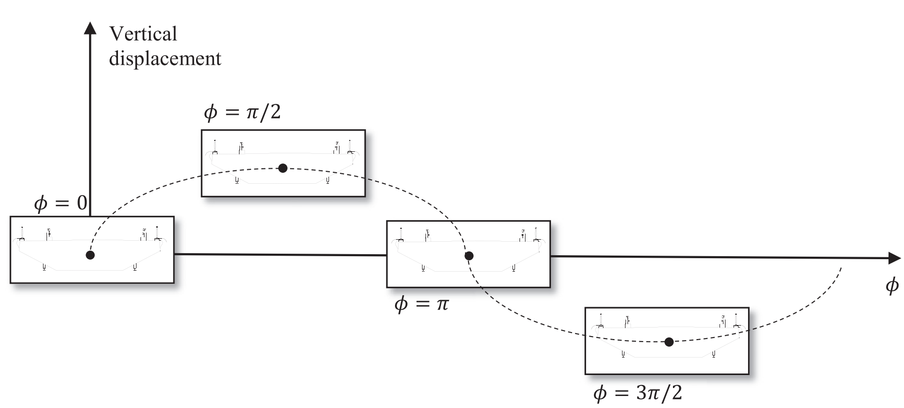

Fig. 10. Phases (ϕ) of heaving motion.

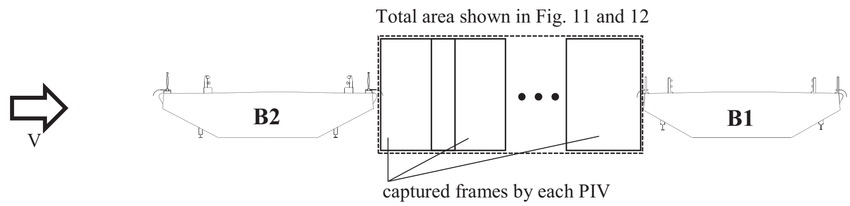

Fig. 11. PIV images.

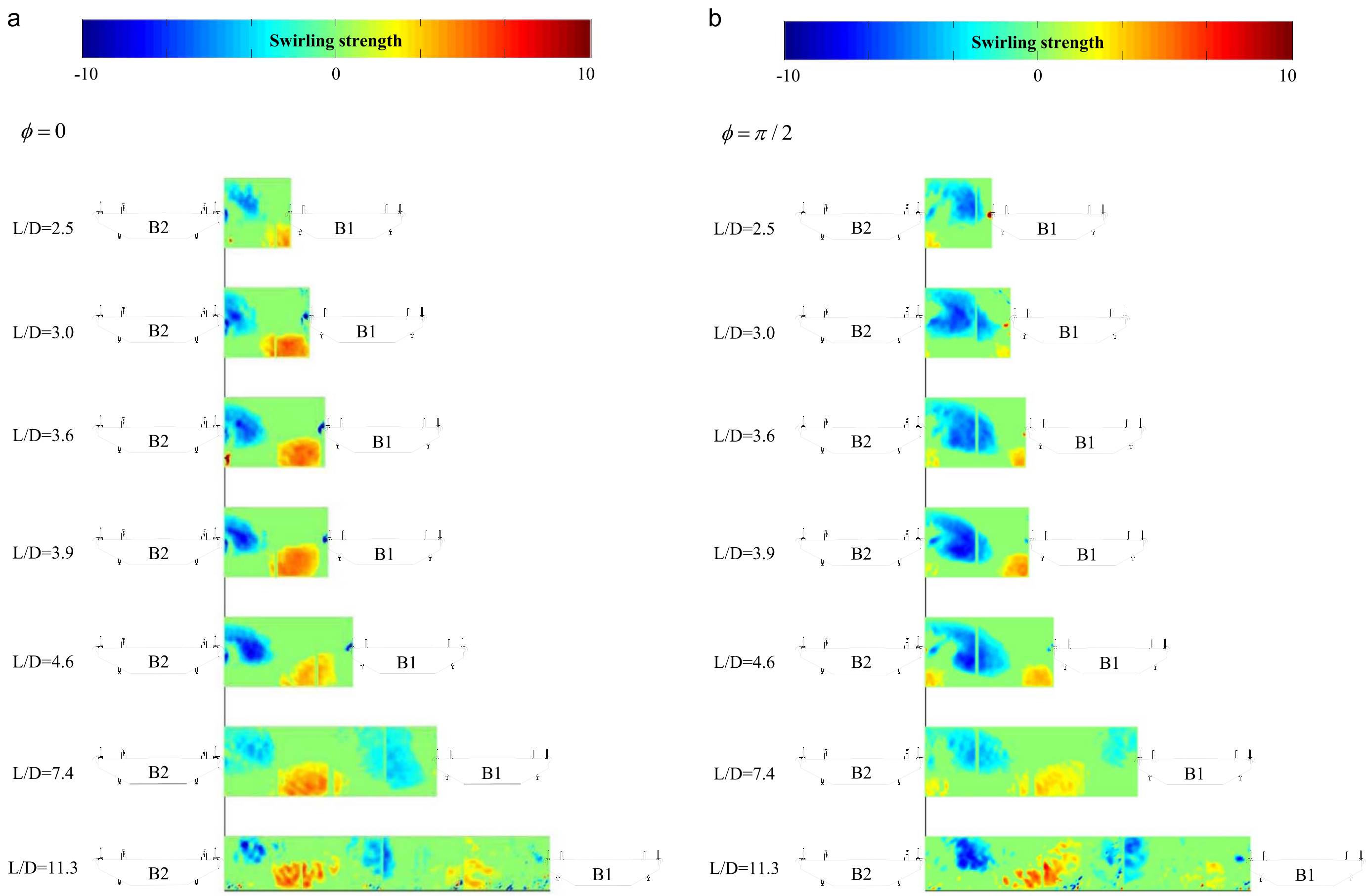

Fig. 12. Swirling strength according to the gap distances at phase of (a) $\phi = 0$ (b) $\phi = \pi/2$ (c) $\phi = \pi$ and (d) $\phi = 3\pi/2$.

VIV in a relatively higher wind velocity than when L/D=2.1, as shown in Fig. 5(c). The range of the lock-in wind velocity of B1 narrowed as the gap distance increased, and finally became enclosed with the lock-in range of B2 at a gap distance of L/D=5.4, as shown in Fig. 5(i). The amplitude of the VIV in B1 was temporarily suppressed to the level of a single deck for this gap distance.

Fig. 6 shows the motional effect of B2 on the VIV in B1. For comparison, the motion of the windward B2 was artificially restrained and the motion of B1 was then monitored, as also tried by Kimura et al. (2008). As shown in Fig. 6(a)-(e), the VIVs in B1 became more active with higher amplitudes as well as wider lock-in wind velocities when the motion of B2 was restrained. The motional effect of B2 on the VIV in B1 was clearly identified at the gap distance of L/D=2.5–3.6 for these bridges in terms of lock-in wind velocities, and this effect weakened as the gap distance increased. However, as the gap distance reached L/D=5.4, the lock-in wind velocities in both decks reached the same range, and the maximum amplitude of B1 was four-fold when both movable cases were compared, as shown in Fig. 6(f).

3.3. *Interactive VIVs for the gap distances of* L/D=*7.4–19.5*

The A-V curves of each deck were similar regardless of the gap distance for this range, as shown in Fig. 5(j)-(n). The motion of B2 was no longer severely affected by the parallel disposition of the two decks and approximated the level of a single deck, as shown in Fig. 5(o).

However, the motions in B1 continued to differ from that of a single deck in terms of amplitude and lock-in range. The time-history of the heaving motion and its spectral density for B1 are demonstrated in Fig. 7 for a gap distance of L/D=7.4. B1 was no longer in a steady-state VIV motion, as shown in Fig. 7(a). The frequency component of the motional history was spread out relative to the natural frequency of B1.

Fig. 8 also shows the spectral density of the vertical wind velocity fluctuation measured behind B2. By passing through B2, the smooth flow in front of B2 became somewhat turbulent. Two peaks corresponding to the motional frequencies for B2 (5.1 Hz) and B1 (5.9 Hz) are shown in Fig. 8. This implies that the wind flows between the two decks were composed of motion-induced components that were tuned

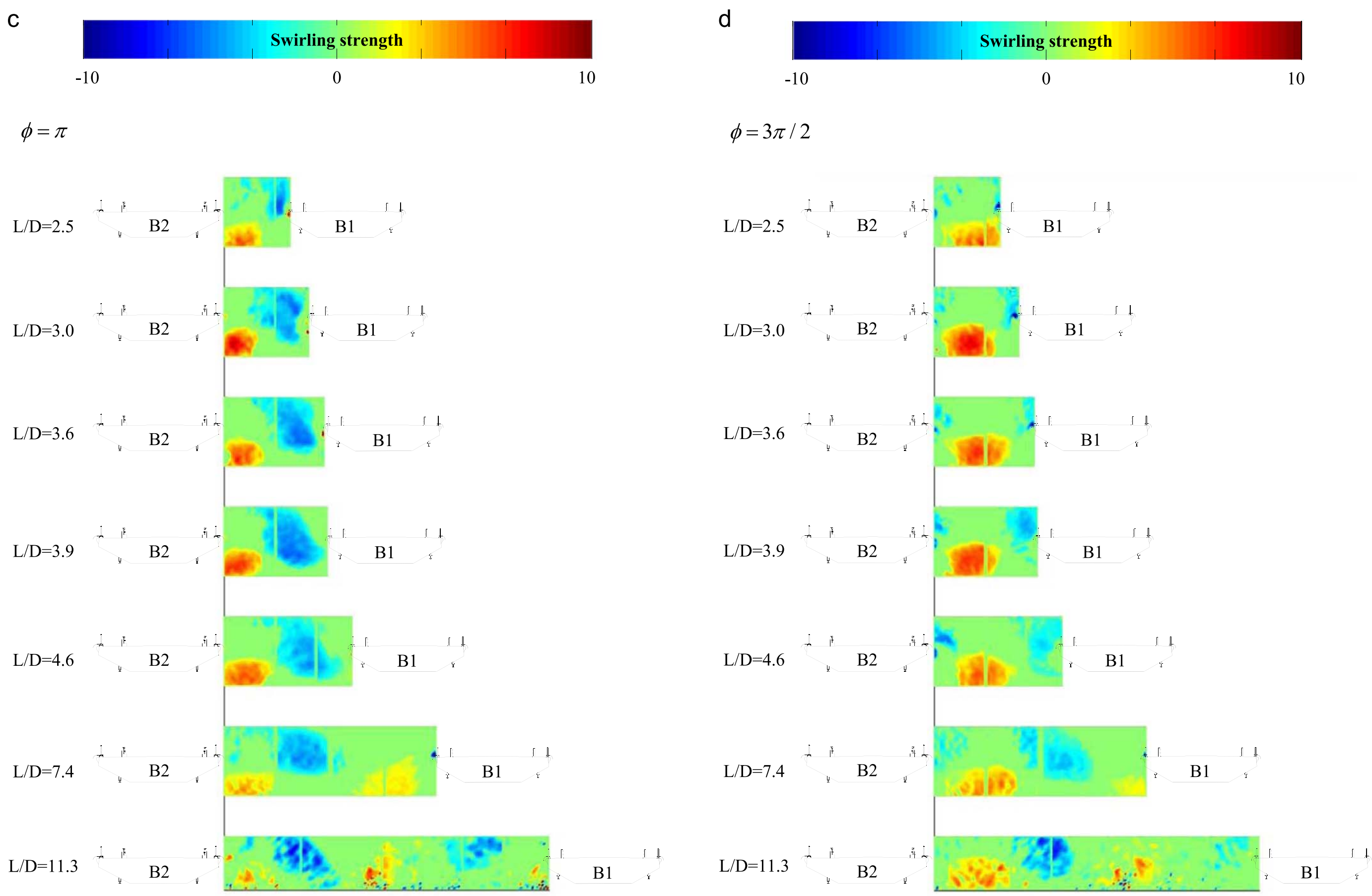

Fig. 12. (*continued*)

to each deck oscillation as well as to the random turbulent components. The turbulent flow induced B1 to oscillate with a peak factor of 2.2–2.7, which belonged somewhere between VIV and buffeting-type vibration.

3.4. Strouhal numbers according to the gap distances

The Strouhal numbers were measured behind B2 with a hot-wire anemometer by fixing both decks in the wind tunnel. Fig. 9 shows the Strouhal numbers estimated from the dominant shedding frequency of vortex trails for the gap distances of L/D=0.5, 0.75, 1.25, 1.5, 2.0, 2.5, 3.0, 3.5, 4.0, 5.0, 6.0, 7.0, 8.0, 9.0, 10.0, 11.0, 12.0, and 13.0. For $L/D < 2.5$, the Strouhal number steadily decreased with an increase in the gap distance. A minimum value of 0.101 was identified at L/D=2.5, and then the Strouhal number gradually increased for $L/D > 2.5$. The value at L/D=13.0 converged to 0.124, which conformed to the value for a single deck of B2.

4. Influence of gap distance on the flow field

PIV tests were performed to demonstrate the change in the flow streams between two decks for different gap distances. A trigger generator, synchronized to the position of B2 during the vertical motion, was utilized to capture repetitive images in the same phases. The phase (ϕ) is defined as shown in Fig. 10. We obtained 200 images for each of 16 phases and applied phase-average analysis (Hussain and Reynolds, 1970). The investigated area of the flow field extended from the leeward edge of B2 to the windward edge of B1, and varied in size according to the gap distance. Since the observed area generally exceeded the size of a single frame of the camera, an entire flow field between two decks was synthesized with divided frames obtained from different camera positions, by overlapping consecutive frames as shown in Fig. 11.

Fig. 12 shows the distribution of the vortex strength along the gap distances for phases of $\phi = 0, \pi/2, \pi$ and $3\pi/2$, respectively. To visualize the vortices, a swirling strength method proposed by Zhou et al. (1999) was utilized. The swirling strength method is based on the decomposition of velocity gradient tensor, $\boldsymbol{D}$, for the Cartesian coordinates (x, y, z). The velocity gradient tensor is written as follows:

$$\boldsymbol{D}=\begin{bmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{bmatrix} \tag{1}$$

where, u, v, and w are the velocity components of the flow particles along the x, y and z directions, respectively. Then, $\boldsymbol{D}$ can be decomposed as follows:

$$\boldsymbol{D}=[\boldsymbol{v_r} \quad \boldsymbol{v_{cr}} \quad \boldsymbol{v_{ci}}]\begin{bmatrix} \lambda_r & & \\ & \lambda_{cr} & \lambda_{ci} \\ & -\lambda_{ci} & \lambda_{cr} \end{bmatrix}[\boldsymbol{v_r} \quad \boldsymbol{v_{cr}} \quad \boldsymbol{v_{ci}}]^{-1} \tag{2}$$

where λ_r and $\lambda_{cr}\pm\lambda_{ci}i$ ($i=\sqrt{-1}$) are the real and the conjugate pair of the complex eigenvalues for the corresponding eigenvectors, $\boldsymbol{v_r}$ and $\boldsymbol{v_{cr}}\pm\boldsymbol{v_{ci}}$, respectively. According to Zhou et al. (1999), the three vectors $[\boldsymbol{v_r} \quad \boldsymbol{v_{cr}} \quad \boldsymbol{v_{ci}}]$ define a local coordinate (y_1, y_2, y_3) system. The local streamlines can be expressed as follows:

$$\begin{aligned} y_1(t)&=C_r\exp\lambda_r t, \\ y_2(t)&=\exp\lambda_{cr}t\,[C_c^1\cos(\lambda_{ci}t)+C_c^2\sin(\lambda_{ci}t)], \\ y_3(t)&=\exp\lambda_{cr}t\,[C_c^2\cos(\lambda_{ci}t)-C_c^1\sin(\lambda_{ci}t)], \end{aligned} \tag{3}$$

where C_r, C_c^1 and C_c^2 are constants for the initial values at $t = 0$. The strength of the local swirling motion is quantified by the imaginary part

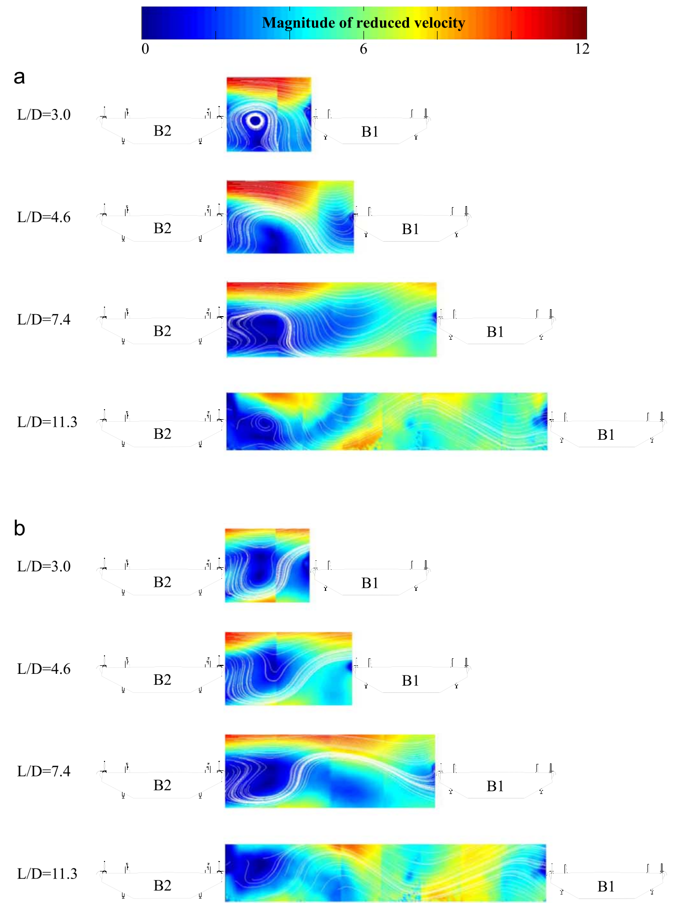

Fig. 13. Magnitude of wind velocity and stream line according to the gap distances at phase of (a) $\phi = \pi/2$ and (b) $\phi = 3\pi/2$.

of the complex eigenvalue, λ_{ci}. Since the PIV test in this study identifies only two-dimensional flows, the velocity gradient tensor can also be represented in two-dimensional form. The determinant for the two-dimensional form of the velocity gradient tensor gives two real eigenvalues or a conjugated pair of complex eigenvalues, $\lambda_{cr}\pm\lambda_{ci}i$, then λ_{ci} is used as the swirling strength.

The direction of vortices can be represented by the vorticity for a two-dimensional incompressible flow as follows:

$$\omega=\frac{\partial v}{\partial x}-\frac{\partial u}{\partial y} \tag{4}$$

which results in the counterclockwise rotation being positive.

The vortices were alternately shed from the top and bottom of the leeward end of B2, as shown in Fig. 12. The total number of vortices generated between two decks depends on the gap distance. In fact, the swirling strength of a vortex was strong when the gap distance ranged from 3.0 to 3.9. For this range, the generated vortex from B2 almost reached B1 during one cycle of motion of B2. When the gap size increased, the shape of the generated vortex was spread out, and the swirling strength became weak. The moving distance of the generated vortex during the one cycle of motion of B2 became shorter than the distance between the two decks. This illustrates the fact that the vortices were well developed when the time duration for the vortices crossing the gap distance approximated the motional period of B2. Accordingly, the formation of magnified vortices was the result of the synchronization between one period of the heaving motion of the windward deck and the time required for the vortex movement between two decks, particularly for gap distances that fell between 2.5 and 4.6. This can be considered another type of the lock-in phenomenon in parallel cable-stayed bridges.

The magnitudes of the vector-sum wind velocity and the streamlines are also shown in Fig. 13. The alternating Ω and U-shaped streamlines, which were reported by Seo et al. (2013), were again well identified between the two decks. When the wind velocity was taken into consideration, the vertical upward and downward flows could be identified with sky-blue color for the gap distances of 3.0 and 4.6, which demonstrates the existence of relatively fast vertical flows between two decks for those gap distances. As the gap distance increased to between 7.4 and 11.3, the vertical flows weakened and the flow pattern was flattened with an increase in horizontal velocity. Simultaneously, the vortices began to spread out into small pieces.

5. Conclusions

The effect of gap distance on interactive VIV for twin cable-stayed bridges was investigated using wind tunnel testing. The gap distances ranged from 0.7 to 19.5 in terms of non-dimensional variables defined as the actual distances divided by the depth of the windward deck. The gap distance was a critical factor in estimating the interactive motions of two bridge decks. The maximum amplitudes of the VIVs and the ranges for the lock-in wind velocities in both bridges changed dramatically according to the gap distance. For small gap distances of 0.7 and 1.3, a VIV could only be identified in the downstream deck. As the gap distance was increased from 2.1 to 5.4, the vortices generated from the leeward edge of the upstream deck were magnified between the two decks resulting in alternating Ω and U-shaped streamlines, which induced observable oscillations in both decks. However, due to the differences in the natural frequencies between the two decks, strong oscillations in both decks were not simultaneously observed. The PIV tests demonstrated that the strength of the vortex reached its maximum level when the time duration for the vortices crossing the gap distance approximated that of the one-motional period of the upstream deck. As the gap distance increased to more than 7.4, the interactive VIVs almost disappeared and only buffeting-type vibrations were observed in the downstream deck due to the disturbed flow by the upstream deck.

Accordingly, we confirmed that the interactive VIV in two parallel cable-stayed bridges is critically affected by gap distances of as much as five to seven times the depth of the deck. This vulnerable gap distance is closely related to the moving distance of one vortex during a one-period oscillation of the deck. The as-built bridges unfortunately had gap distances that were within the range of an interactive VIV region, and this contributed to the VIV observed in the target bridges.

This study was focused only on the effect that gap distance had on the VIV in two parallel cable-stayed bridges. For more insight into the complex interactions between two decks, the parameters such as the relative differences in natural frequencies and the outer shapes of two decks are recommended for further studies. The effect of reversed wind direction could also be investigated during those studies by switching the relative natural frequencies and/or the outer shapes of the two bridges in a wind tunnel.

Acknowledgements

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